

Lexical Grammar

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1 Ambiguity

Ambiguity is recognized between some of the regexes listed below. This ambiguity is solved in the lexer by taking the longest match or through regex precedence. The longest match is to pick the regex that matches the most characters (greedy matching), while the precedence resolution is to define a precedence of regexes such that in the case where two regexes match the same span one is preferred.

1.1 Examples

Some examples of ambiguous matches include:

- "else" matches both $\langle \textit{else} \rangle$ and $\langle \textit{ident} \rangle$. Solved with precedence.
- "1.2e3" matches as either $\langle \textit{float_const} \rangle$ followed by an $\langle \textit{ident} \rangle$ or as a $\langle \textit{scientific} \rangle$. Solved with longest match.
- "r\"abc\"" matches as either an $\langle \textit{ident} \rangle$ followed by a $\langle \textit{string_literal} \rangle$ or as just a $\langle \textit{string_literal} \rangle$. Solved with longest match.

2 Regex

2.1 Substitutions

L	$:=$	$[a - z]$	(1)
U	$:=$	$[A - Z]$	(2)
D	$:=$	$[0 - 9]$	(3)
H	$:=$	$[a - fA - F0 - 9]$	(4)
O	$:=$	$[0 - 7]$	(5)
B	$:=$	$[01]$	(6)

2.2 Identifiers, Literals, and Comments

$_ \{0, 2\}L(? : L U D _)*$	$\rightarrow$	IDENT	(7)
$U(? : L U D)*$	$\rightarrow$	TYPE_IDENT	(8)
$[_ + _]?D+$	$\rightarrow$	INTEGER_LITERAL	(9)
$0[\text{xX}]H\{1, 16\}$	$\rightarrow$	NUMERIC_LITERAL	Hex (10)
$0[\text{oO}]O\{1, 22\}$	$\rightarrow$	NUMERIC_LITERAL	Octal (11)
$0[\text{bB}]B\{1, 64\}$	$\rightarrow$	NUMERIC_LITERAL	Binary (12)
$([_ + _]?(? : D + (? : _.D*)?[_.D+]))[eE]([_ + _]?D+)$	$\rightarrow$	SCIENTIFIC_LITERAL	(13)
$[_ + _]?(? : D + _.D*[_.D+)$	$\rightarrow$	FLOAT_LITERAL	(14)
$(\text{r}?)^* ((? : [\text{^} \text{"} \backslash] \backslash \text{"})^*)^*$	$\rightarrow$	STRING_LITERAL	(15)
$\#.\text{*}\$$	$\rightarrow$	COMMENT	(16)

2.3 Symbols

,	→	COMMA	(17)
;	→	SEMI	(18)
(	→	LEFT_PAREN	(19)
)	→	RIGHT_PAREN	(20)
[	→	LEFT_SQUARE	(21)
]	→	RIGHT_SQUARE	(22)
{	→	LEFT_CURLY	(23)
}	→	RIGHT_CURLY	(24)

2.4 Keywords

and	→	LOGICAL_AND	(25)
bool	→	BOOL	(26)
break	→	BREAK	(27)
class	→	CLASS	(28)
elif	→	ELIF	(29)
else	→	ELSE	(30)
enum	→	ENUM	(31)
exit	→	EXIT	(32)
false	→	FALSE	(33)
float32	→	FLOAT32	(34)
float64	→	FLOAT64	(35)
for	→	FOR	(36)
func	→	FUNCTION	(37)
if	→	IF	(38)
iface	→	INTERFACE	(39)
in	→	IN	(40)
int16	→	INT16	(41)
int32	→	INT32	(42)
int64	→	INT64	(43)
int8	→	INT8	(44)
mut	→	MUTABLE	(45)
or	→	LOGICAL_OR	(46)
private	→	PRIVATE	(47)
public	→	PUBLIC	(48)
string	→	STRING	(49)
true	→	TRUE	(50)
uint16	→	UINT16	(51)
uint32	→	UINT32	(52)
uint64	→	UINT64	(53)
uint8	→	UINT8	(54)
while	→	WHILE	(55)

2.5 Operators

$\wedge\wedge=$	→	EXPONENT_ASSIGN	(56)
$\wedge\wedge$	→	EXPONENT	(57)
$\wedge=$	→	XOR_ASSIGN	(58)
$\wedge$	→	XOR	(59)
$\% =$	→	MOD_ASSIGN	(60)
$\%$	→	MOD	(61)
$* =$	→	MUL_ASSIGN	(62)
$*$	→	MUL	(63)
$// =$	→	INTEGER_DIV_ASSIGN	(64)
$//$	→	INTEGER_DIV	(65)
$/ =$	→	DIV_ASSIGN	(66)
$/$	→	DIV	(67)
$++$	→	INC	(68)
$+ =$	→	ADD_ASSIGN	(69)
$+$	→	ADD	(70)
$--$	→	DEC	(71)
$- =$	→	SUB_ASSIGN	(72)
$-$	→	SUB	(73)
$<<$	→	SHIFT_LEFT	(74)
$<=$	→	LE_OP	(75)
$<$	→	LT_OP	(76)
$>>$	→	SHIFT_RIGHT	(77)
$>=$	→	GE_OP	(78)
$>$	→	GT_OP	(79)
$!=$	→	NE_OP	(80)
$==$	→	EQ_OP	(81)
$=$	→	ASSIGN	(82)
$\& =$	→	AND_ASSIGN	(83)
$\&$	→	AND	(84)
$ =$	→	OR_ASSIGN	(85)
$ $	→	OR	(86)
$?$	→	TERNARY	(87)
$.$	→	DOT	(88)

3 Backus-Naur Form

$\langle \text{ident} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{ident?} \rangle$	(89)
		$ \text{"_"} \langle \text{lower} \rangle \langle \text{ident?} \rangle$	(90)
		$ \text{"_"} \text{"_"} \langle \text{lower} \rangle \langle \text{ident?} \rangle$	(91)
$\langle \text{ident?} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{ident?} \rangle$	(92)
		$ \langle \text{upper} \rangle \langle \text{ident?} \rangle$	(93)
		$ \langle \text{digit} \rangle \langle \text{ident?} \rangle$	(94)
		$ \text{"_"} \langle \text{ident?} \rangle$	(95)
		$ \varepsilon$	(96)
$\langle \text{type_ident} \rangle$	$::=$	$\langle \text{upper} \rangle \langle \text{type_ident?} \rangle$	(97)
$\langle \text{type_ident?} \rangle$	$::=$	$\langle \text{type_ident} \rangle$	(98)
		$ \langle \text{lower} \rangle \langle \text{type_ident?} \rangle$	(99)
		$ \langle \text{digit} \rangle \langle \text{type_ident?} \rangle$	(100)
		$ \varepsilon$	(101)
$\langle \text{float_const} \rangle$	$::=$	$\langle \text{scientific} \rangle$	(102)
		$ \langle \text{mantissa} \rangle$	(103)
		$ \langle \text{decimal_const} \rangle$	(104)
		$ \langle \text{hex_const} \rangle$	(105)
		$ \langle \text{octal_const} \rangle$	(106)
		$ \langle \text{binary_const} \rangle$	(107)
$\langle \text{integer_const} \rangle$	$::=$	$\langle \text{decimal_const} \rangle$	(108)
		$ \langle \text{hex_const} \rangle$	(109)
		$ \langle \text{octal_const} \rangle$	(110)
		$ \langle \text{binary_const} \rangle$	(111)
$\langle \text{comment} \rangle$	$::=$	$\text{"\#"} \langle \text{comment_text} \rangle$	(112)
$\langle \text{comment_text} \rangle$	$::=$	$\langle \text{comment_char} \rangle \langle \text{comment_text} \rangle$	(113)
$\langle \text{comment_char} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{upper} \rangle \langle \text{digit} \rangle \langle \text{symbol} \rangle \langle \text{non_newline} \rangle$	(114)
$\langle \text{string_literal} \rangle$	$::=$	$\langle \text{r?} \rangle \langle \text{quote} \rangle \langle \text{content} \rangle \langle \text{quote} \rangle$	(115)
$\langle \text{content} \rangle$	$::=$	$\langle \text{char} \rangle \langle \text{content} \rangle \varepsilon$	(116)
$\langle \text{char} \rangle$	$::=$	$\langle \text{escaped_quote} \rangle \langle \text{normal_symbol} \rangle$	(117)
$\langle \text{scientific} \rangle$	$::=$	$\langle \text{mantissa} \rangle \langle \text{exponent} \rangle$	(118)
		$ \langle \text{decimal_const} \rangle \langle \text{exponent} \rangle$	(119)
$\langle \text{mantissa} \rangle$	$::=$	$\langle \text{sign?} \rangle \langle \text{digits} \rangle \text{"."} \langle \text{digits?} \rangle$	(120)
		$ \langle \text{sign?} \rangle \text{"."} \langle \text{digits} \rangle$	(121)
$\langle \text{exponent} \rangle$	$::=$	$\text{"e"} \langle \text{decimal_const} \rangle$	(122)
$\langle \text{decimal_const} \rangle$	$::=$	$\langle \text{sign?} \rangle \langle \text{digits} \rangle$	(123)
$\langle \text{hex_const} \rangle$	$::=$	$\text{"0"} \text{"x"} \langle \text{hs} \rangle$	(124)
		$ \text{"0"} \text{"X"} \langle \text{hs} \rangle$	(125)
$\langle \text{octal_const} \rangle$	$::=$	$\text{"0"} \text{"o"} \langle \text{os} \rangle$	(126)
		$ \text{"0"} \text{"O"} \langle \text{os} \rangle$	(127)
$\langle \text{binary_const} \rangle$	$::=$	$\text{"0"} \text{"b"} \langle \text{bs} \rangle$	(128)
		$ \text{"0"} \text{"B"} \langle \text{bs} \rangle$	(129)

<code>< and ></code>	<code>::=</code>	<code>"a""n""d"</code>	(130)
<code>< bool ></code>	<code>::=</code>	<code>"b""o""o""l"</code>	(131)
<code>< break ></code>	<code>::=</code>	<code>"b""r""e""a""k"</code>	(132)
<code>< class ></code>	<code>::=</code>	<code>"c""l""a""s""s"</code>	(133)
<code>< elif ></code>	<code>::=</code>	<code>"e""l""i""f"</code>	(134)
<code>< else ></code>	<code>::=</code>	<code>"e""l""s""e"</code>	(135)
<code>< enum ></code>	<code>::=</code>	<code>"e""n""u""m"</code>	(136)
<code>< exit ></code>	<code>::=</code>	<code>"e""x""i""t"</code>	(137)
<code>< false ></code>	<code>::=</code>	<code>"f""a""l""s""e"</code>	(138)
<code>< float32 ></code>	<code>::=</code>	<code>"f""l""o""a""t""3""2"</code>	(139)
<code>< float64 ></code>	<code>::=</code>	<code>"f""l""o""a""t""6""4"</code>	(140)
<code>< for ></code>	<code>::=</code>	<code>"f""o""r"</code>	(141)
<code>< func ></code>	<code>::=</code>	<code>"f""u""n""c"</code>	(142)
<code>< if ></code>	<code>::=</code>	<code>"i""f"</code>	(143)
<code>< iface ></code>	<code>::=</code>	<code>"i""f""a""c""e"</code>	(144)
<code>< in ></code>	<code>::=</code>	<code>"i""n"</code>	(145)
<code>< int16 ></code>	<code>::=</code>	<code>"i""n""t""1""6"</code>	(146)
<code>< int32 ></code>	<code>::=</code>	<code>"i""n""t""3""2"</code>	(147)
<code>< int64 ></code>	<code>::=</code>	<code>"i""n""t""6""4"</code>	(148)
<code>< int8 ></code>	<code>::=</code>	<code>"i""n""t""8"</code>	(149)
<code>< mut ></code>	<code>::=</code>	<code>"m""u""t"</code>	(150)
<code>< or ></code>	<code>::=</code>	<code>"o""r"</code>	(151)
<code>< private ></code>	<code>::=</code>	<code>"p""r""i""v""a""t""e"</code>	(152)
<code>< public ></code>	<code>::=</code>	<code>"p""u""b""l""i""c"</code>	(153)
<code>< string ></code>	<code>::=</code>	<code>"s""t""r""i""n""g"</code>	(154)
<code>< true ></code>	<code>::=</code>	<code>"t""r""u""e"</code>	(155)
<code>< uint16 ></code>	<code>::=</code>	<code>"u""i""n""t""1""6"</code>	(156)
<code>< uint32 ></code>	<code>::=</code>	<code>"u""i""n""t""3""2"</code>	(157)
<code>< uint64 ></code>	<code>::=</code>	<code>"u""i""n""t""6""4"</code>	(158)
<code>< uint8 ></code>	<code>::=</code>	<code>"u""i""n""t""8"</code>	(159)
<code>< while ></code>	<code>::=</code>	<code>"w""h""i""l""e"</code>	(160)

$\langle \textit{exponent_assign} \rangle$	::=	" ^ " " ^ " " = "	(161)
$\langle \textit{exponent} \rangle$	::=	" ^ " " ^ "	(162)
$\langle \textit{xor_assign} \rangle$	::=	" ^ " " = "	(163)
$\langle \textit{xor} \rangle$	::=	" ^ "	(164)
$\langle \textit{mod_assign} \rangle$	::=	" % " " = "	(165)
$\langle \textit{mod} \rangle$	::=	" % "	(166)
$\langle \textit{mul_assign} \rangle$	::=	" * " " = "	(167)
$\langle \textit{mul} \rangle$	::=	" * "	(168)
$\langle \textit{integer_div_assign} \rangle$	::=	" / " " / " " = "	(169)
$\langle \textit{integer_div} \rangle$	::=	" / " " / "	(170)
$\langle \textit{div_assign} \rangle$	::=	" / " " = "	(171)
$\langle \textit{div} \rangle$	::=	" / "	(172)
$\langle \textit{inc} \rangle$	::=	" + " " + "	(173)
$\langle \textit{add_assign} \rangle$	::=	" + " " = "	(174)
$\langle \textit{add} \rangle$	::=	" + "	(175)
$\langle \textit{dec} \rangle$	::=	" - " " - "	(176)
$\langle \textit{sub_assign} \rangle$	::=	" - " " = "	(177)
$\langle \textit{sub} \rangle$	::=	" - "	(178)
$\langle \textit{shift_left} \rangle$	::=	" < " " < "	(179)
$\langle \textit{le_op} \rangle$	::=	" < " " = "	(180)
$\langle \textit{lt_op} \rangle$	::=	" < "	(181)
$\langle \textit{shift_right} \rangle$	::=	" > " " > "	(182)
$\langle \textit{ge_op} \rangle$	::=	" > " " = "	(183)
$\langle \textit{gt_op} \rangle$	::=	" > "	(184)
$\langle \textit{ne_op} \rangle$	::=	" ! " " = "	(185)
$\langle \textit{eq_op} \rangle$	::=	" = " " = "	(186)
$\langle \textit{assign} \rangle$	::=	" = "	(187)
$\langle \textit{and_assign} \rangle$	::=	" & " " = "	(188)
$\langle \textit{and} \rangle$	::=	" & "	(189)
$\langle \textit{or_assign} \rangle$	::=	" " " = "	(190)
$\langle \textit{or} \rangle$	::=	" "	(191)
$\langle \textit{ternary} \rangle$	::=	" ? "	(192)
$\langle \textit{dot} \rangle$	::=	" . "	(193)

$\langle \text{digits?} \rangle$	$::= \langle \text{digits} \rangle \mid \varepsilon$	(194)
$\langle \text{digits} \rangle$	$::= \langle \text{digit} \rangle \mid \langle \text{digit} \rangle \langle \text{digits} \rangle$	(195)
$\langle \text{sign?} \rangle$	$::= \langle \text{sign} \rangle \mid \varepsilon$	(196)
$\langle \text{sign} \rangle$	$::= "+" \mid "-"$	(197)
$\langle \text{lower} \rangle$	$::= "a" \mid "b" \mid "c" \mid "d" \mid "e" \mid "f" \mid "g" \mid "h" \mid "i" \mid "j" \mid "k" \mid "l" \mid "m" \mid$ $"n" \mid "o" \mid "p" \mid "q" \mid "r" \mid "s" \mid "t" \mid "u" \mid "v" \mid "w" \mid "x" \mid "y" \mid "z"$	(198)
$\langle \text{upper} \rangle$	$::= "A" \mid "B" \mid "C" \mid "D" \mid "E" \mid "F" \mid "G" \mid "H" \mid "I" \mid "J" \mid "K" \mid "L" \mid "M" \mid$ $"N" \mid "O" \mid "P" \mid "Q" \mid "R" \mid "S" \mid "T" \mid "U" \mid "V" \mid "W" \mid "X" \mid "Y" \mid "Z"$	(199)
$\langle \text{digit} \rangle$	$::= "0" \mid "1" \mid "2" \mid "3" \mid "4" \mid "5" \mid "6" \mid "7" \mid "8" \mid "9"$	(200)
$\langle \text{hex} \rangle$	$::= "0" \mid "1" \mid "2" \mid "3" \mid "4" \mid "5" \mid "6" \mid "7" \mid "8" \mid "9" \mid$ $"a" \mid "b" \mid "c" \mid "d" \mid "e" \mid "f" \mid "A" \mid "B" \mid "C" \mid "D" \mid "E" \mid "F"$	(201)
$\langle \text{octal} \rangle$	$::= "0" \mid "1" \mid "2" \mid "3" \mid "4" \mid "5" \mid "6" \mid "7"$	(202)
$\langle \text{binary} \rangle$	$::= "0" \mid "1"$	(203)
$\langle \text{symbols} \rangle$	$::= \langle \text{normal_symbol} \rangle \mid \langle \text{quote} \rangle \mid \langle \text{backslash} \rangle$	(204)
$\langle \text{normal_symbol} \rangle$	$::= "!" \mid "\#" \mid "$" \mid "\%" \mid "\&" \mid "'" \mid "(" \mid ")" \mid "*" \mid "+" \mid "," \mid "-" \mid "." \mid$ $"/" \mid ":" \mid ";" \mid "<" \mid "=" \mid ">" \mid "?" \mid "@" \mid "[" \mid "]" \mid "[" \mid "]" \mid "-" \mid "_" \mid "$ $"\{" \mid "\}" \mid "\}" \mid "$	(205)
$\langle \text{whitespace} \rangle$	$::= \langle \text{non_newline} \rangle \mid \langle \text{newline} \rangle$	(206)
$\langle \text{non_newline} \rangle$	$::= "" \mid "\backslash t" \mid "\backslash r"$	(207)
$\langle \text{newline} \rangle$	$::= "\backslash n"$	(208)
$\langle \text{r?} \rangle$	$::= "r" \mid \varepsilon$	(209)
$\langle \text{escaped_quote} \rangle$	$::= \langle \text{backslash} \rangle \langle \text{quote} \rangle$	(210)
$\langle \text{backslash} \rangle$	$::= "\backslash"$	(211)
$\langle \text{quote} \rangle$	$::= ""$	(212)
		(213)

$\langle hs \rangle$	$::=$	$\langle hex \rangle$	(214)
		$ \langle hex \rangle \langle hs1 \rangle$	(215)
$\langle hs1 \rangle$	$::=$	$\langle hex \rangle$	(216)
		$ \langle hex \rangle \langle hs2 \rangle$	(217)
$\langle hs2 \rangle$	$::=$	$\langle hex \rangle$	(218)
		$ \langle hex \rangle \langle hs3 \rangle$	(219)
$\langle hs3 \rangle$	$::=$	$\langle hex \rangle$	(220)
		$ \langle hex \rangle \langle hs4 \rangle$	(221)
$\langle hs4 \rangle$	$::=$	$\langle hex \rangle$	(222)
		$ \langle hex \rangle \langle hs5 \rangle$	(223)
$\langle hs5 \rangle$	$::=$	$\langle hex \rangle$	(224)
		$ \langle hex \rangle \langle hs6 \rangle$	(225)
$\langle hs6 \rangle$	$::=$	$\langle hex \rangle$	(226)
		$ \langle hex \rangle \langle hs7 \rangle$	(227)
$\langle hs7 \rangle$	$::=$	$\langle hex \rangle$	(228)
		$ \langle hex \rangle \langle hs8 \rangle$	(229)
$\langle hs8 \rangle$	$::=$	$\langle hex \rangle$	(230)
		$ \langle hex \rangle \langle hs9 \rangle$	(231)
$\langle hs9 \rangle$	$::=$	$\langle hex \rangle$	(232)
		$ \langle hex \rangle \langle hs10 \rangle$	(233)
$\langle hs10 \rangle$	$::=$	$\langle hex \rangle$	(234)
		$ \langle hex \rangle \langle hs11 \rangle$	(235)
$\langle hs11 \rangle$	$::=$	$\langle hex \rangle$	(236)
		$ \langle hex \rangle \langle hs12 \rangle$	(237)
$\langle hs12 \rangle$	$::=$	$\langle hex \rangle$	(238)
		$ \langle hex \rangle \langle hs13 \rangle$	(239)
$\langle hs13 \rangle$	$::=$	$\langle hex \rangle$	(240)
		$ \langle hex \rangle \langle hs14 \rangle$	(241)
$\langle hs14 \rangle$	$::=$	$\langle hex \rangle$	(242)
		$ \langle hex \rangle \langle hs15 \rangle$	(243)
$\langle hs15 \rangle$	$::=$	$\langle hex \rangle$	(244)

$\langle os \rangle$	$::=$	$\langle octal \rangle$	(245)
		$ \langle octal \rangle \langle os1 \rangle$	(246)
$\langle os1 \rangle$	$::=$	$\langle octal \rangle$	(247)
		$ \langle octal \rangle \langle os2 \rangle$	(248)
$\langle os2 \rangle$	$::=$	$\langle octal \rangle$	(249)
		$ \langle octal \rangle \langle os3 \rangle$	(250)
$\langle os3 \rangle$	$::=$	$\langle octal \rangle$	(251)
		$ \langle octal \rangle \langle os4 \rangle$	(252)
$\langle os4 \rangle$	$::=$	$\langle octal \rangle$	(253)
		$ \langle octal \rangle \langle os5 \rangle$	(254)
$\langle os5 \rangle$	$::=$	$\langle octal \rangle$	(255)
		$ \langle octal \rangle \langle os6 \rangle$	(256)
$\langle os6 \rangle$	$::=$	$\langle octal \rangle$	(257)
		$ \langle octal \rangle \langle os7 \rangle$	(258)
$\langle os7 \rangle$	$::=$	$\langle octal \rangle$	(259)
		$ \langle octal \rangle \langle os8 \rangle$	(260)
$\langle os8 \rangle$	$::=$	$\langle octal \rangle$	(261)
		$ \langle octal \rangle \langle os9 \rangle$	(262)
$\langle os9 \rangle$	$::=$	$\langle octal \rangle$	(263)
		$ \langle octal \rangle \langle os10 \rangle$	(264)
$\langle os10 \rangle$	$::=$	$\langle octal \rangle$	(265)
		$ \langle octal \rangle \langle os11 \rangle$	(266)
$\langle os11 \rangle$	$::=$	$\langle octal \rangle$	(267)
		$ \langle octal \rangle \langle os12 \rangle$	(268)
$\langle os12 \rangle$	$::=$	$\langle octal \rangle$	(269)
		$ \langle octal \rangle \langle os13 \rangle$	(270)
$\langle os13 \rangle$	$::=$	$\langle octal \rangle$	(271)
		$ \langle octal \rangle \langle os14 \rangle$	(272)
$\langle os14 \rangle$	$::=$	$\langle octal \rangle$	(273)
		$ \langle octal \rangle \langle os15 \rangle$	(274)
$\langle os15 \rangle$	$::=$	$\langle octal \rangle$	(275)
		$ \langle octal \rangle \langle os16 \rangle$	(276)
$\langle os16 \rangle$	$::=$	$\langle octal \rangle$	(277)
		$ \langle octal \rangle \langle os17 \rangle$	(278)
$\langle os17 \rangle$	$::=$	$\langle octal \rangle$	(279)
		$ \langle octal \rangle \langle os18 \rangle$	(280)
$\langle os18 \rangle$	$::=$	$\langle octal \rangle$	(281)
		$ \langle octal \rangle \langle os19 \rangle$	(282)
$\langle os19 \rangle$	$::=$	$\langle octal \rangle$	(283)
		$ \langle octal \rangle \langle os20 \rangle$	(284)
$\langle os20 \rangle$	$::=$	$\langle octal \rangle$	(285)
		$ \langle octal \rangle \langle os21 \rangle$	(286)
$\langle os21 \rangle$	$::=$	$\langle octal \rangle$	(287)

$\langle bs \rangle$	::=	$\langle binary \rangle$	(288)
		$ \langle binary \rangle \langle bs1 \rangle$	(289)
$\langle bs1 \rangle$	::=	$\langle binary \rangle$	(290)
		$ \langle binary \rangle \langle bs2 \rangle$	(291)
$\langle bs2 \rangle$	::=	$\langle binary \rangle$	(292)
		$ \langle binary \rangle \langle bs3 \rangle$	(293)
$\langle bs3 \rangle$	::=	$\langle binary \rangle$	(294)
		$ \langle binary \rangle \langle bs4 \rangle$	(295)
$\langle bs4 \rangle$	::=	$\langle binary \rangle$	(296)
		$ \langle binary \rangle \langle bs5 \rangle$	(297)
$\langle bs5 \rangle$	::=	$\langle binary \rangle$	(298)
		$ \langle binary \rangle \langle bs6 \rangle$	(299)
$\langle bs6 \rangle$	::=	$\langle binary \rangle$	(300)
		$ \langle binary \rangle \langle bs7 \rangle$	(301)

$\langle bs7 \rangle$	$::=$	$\langle binary \rangle$	(302)
		$ \langle binary \rangle \langle bs8 \rangle$	(303)
$\langle bs8 \rangle$	$::=$	$\langle binary \rangle$	(304)
		$ \langle binary \rangle \langle bs9 \rangle$	(305)
$\langle bs9 \rangle$	$::=$	$\langle binary \rangle$	(306)
		$ \langle binary \rangle \langle bs10 \rangle$	(307)
$\langle bs10 \rangle$	$::=$	$\langle binary \rangle$	(308)
		$ \langle binary \rangle \langle bs11 \rangle$	(309)
$\langle bs11 \rangle$	$::=$	$\langle binary \rangle$	(310)
		$ \langle binary \rangle \langle bs12 \rangle$	(311)
$\langle bs12 \rangle$	$::=$	$\langle binary \rangle$	(312)
		$ \langle binary \rangle \langle bs13 \rangle$	(313)
$\langle bs13 \rangle$	$::=$	$\langle binary \rangle$	(314)
		$ \langle binary \rangle \langle bs14 \rangle$	(315)
$\langle bs14 \rangle$	$::=$	$\langle binary \rangle$	(316)
		$ \langle binary \rangle \langle bs15 \rangle$	(317)
$\langle bs15 \rangle$	$::=$	$\langle binary \rangle$	(318)
		$ \langle binary \rangle \langle bs16 \rangle$	(319)
$\langle bs16 \rangle$	$::=$	$\langle binary \rangle$	(320)
		$ \langle binary \rangle \langle bs17 \rangle$	(321)
$\langle bs17 \rangle$	$::=$	$\langle binary \rangle$	(322)
		$ \langle binary \rangle \langle bs18 \rangle$	(323)
$\langle bs18 \rangle$	$::=$	$\langle binary \rangle$	(324)
		$ \langle binary \rangle \langle bs19 \rangle$	(325)
$\langle bs19 \rangle$	$::=$	$\langle binary \rangle$	(326)
		$ \langle binary \rangle \langle bs20 \rangle$	(327)
$\langle bs20 \rangle$	$::=$	$\langle binary \rangle$	(328)
		$ \langle binary \rangle \langle bs21 \rangle$	(329)
$\langle bs21 \rangle$	$::=$	$\langle binary \rangle$	(330)
		$ \langle binary \rangle \langle bs22 \rangle$	(331)
$\langle bs22 \rangle$	$::=$	$\langle binary \rangle$	(332)
		$ \langle binary \rangle \langle bs23 \rangle$	(333)
$\langle bs23 \rangle$	$::=$	$\langle binary \rangle$	(334)
		$ \langle binary \rangle \langle bs24 \rangle$	(335)
$\langle bs24 \rangle$	$::=$	$\langle binary \rangle$	(336)
		$ \langle binary \rangle \langle bs25 \rangle$	(337)
$\langle bs25 \rangle$	$::=$	$\langle binary \rangle$	(338)
		$ \langle binary \rangle \langle bs26 \rangle$	(339)
$\langle bs26 \rangle$	$::=$	$\langle binary \rangle$	(340)
		$ \langle binary \rangle \langle bs27 \rangle$	(341)

$\langle bs27 \rangle$	$::=$	$\langle binary \rangle$	(342)
		$ \langle binary \rangle \langle bs28 \rangle$	(343)
$\langle bs28 \rangle$	$::=$	$\langle binary \rangle$	(344)
		$ \langle binary \rangle \langle bs29 \rangle$	(345)
$\langle bs29 \rangle$	$::=$	$\langle binary \rangle$	(346)
		$ \langle binary \rangle \langle bs30 \rangle$	(347)
$\langle bs30 \rangle$	$::=$	$\langle binary \rangle$	(348)
		$ \langle binary \rangle \langle bs31 \rangle$	(349)
$\langle bs31 \rangle$	$::=$	$\langle binary \rangle$	(350)
		$ \langle binary \rangle \langle bs32 \rangle$	(351)
$\langle bs32 \rangle$	$::=$	$\langle binary \rangle$	(352)
		$ \langle binary \rangle \langle bs33 \rangle$	(353)
$\langle bs33 \rangle$	$::=$	$\langle binary \rangle$	(354)
		$ \langle binary \rangle \langle bs34 \rangle$	(355)
$\langle bs34 \rangle$	$::=$	$\langle binary \rangle$	(356)
		$ \langle binary \rangle \langle bs35 \rangle$	(357)
$\langle bs35 \rangle$	$::=$	$\langle binary \rangle$	(358)
		$ \langle binary \rangle \langle bs36 \rangle$	(359)
$\langle bs36 \rangle$	$::=$	$\langle binary \rangle$	(360)
		$ \langle binary \rangle \langle bs37 \rangle$	(361)
$\langle bs37 \rangle$	$::=$	$\langle binary \rangle$	(362)
		$ \langle binary \rangle \langle bs38 \rangle$	(363)
$\langle bs38 \rangle$	$::=$	$\langle binary \rangle$	(364)
		$ \langle binary \rangle \langle bs39 \rangle$	(365)
$\langle bs39 \rangle$	$::=$	$\langle binary \rangle$	(366)
		$ \langle binary \rangle \langle bs40 \rangle$	(367)
$\langle bs40 \rangle$	$::=$	$\langle binary \rangle$	(368)
		$ \langle binary \rangle \langle bs41 \rangle$	(369)
$\langle bs41 \rangle$	$::=$	$\langle binary \rangle$	(370)
		$ \langle binary \rangle \langle bs42 \rangle$	(371)
$\langle bs42 \rangle$	$::=$	$\langle binary \rangle$	(372)
		$ \langle binary \rangle \langle bs43 \rangle$	(373)
$\langle bs43 \rangle$	$::=$	$\langle binary \rangle$	(374)
		$ \langle binary \rangle \langle bs44 \rangle$	(375)
$\langle bs44 \rangle$	$::=$	$\langle binary \rangle$	(376)
		$ \langle binary \rangle \langle bs45 \rangle$	(377)
$\langle bs45 \rangle$	$::=$	$\langle binary \rangle$	(378)
		$ \langle binary \rangle \langle bs46 \rangle$	(379)
$\langle bs46 \rangle$	$::=$	$\langle binary \rangle$	(380)
		$ \langle binary \rangle \langle bs47 \rangle$	(381)

$\langle bs47 \rangle$	$::=$	$\langle binary \rangle$	(382)
		$ \langle binary \rangle \langle bs48 \rangle$	(383)
$\langle bs48 \rangle$	$::=$	$\langle binary \rangle$	(384)
		$ \langle binary \rangle \langle bs49 \rangle$	(385)
$\langle bs49 \rangle$	$::=$	$\langle binary \rangle$	(386)
		$ \langle binary \rangle \langle bs50 \rangle$	(387)
$\langle bs50 \rangle$	$::=$	$\langle binary \rangle$	(388)
		$ \langle binary \rangle \langle bs51 \rangle$	(389)
$\langle bs51 \rangle$	$::=$	$\langle binary \rangle$	(390)
		$ \langle binary \rangle \langle bs52 \rangle$	(391)
$\langle bs52 \rangle$	$::=$	$\langle binary \rangle$	(392)
		$ \langle binary \rangle \langle bs53 \rangle$	(393)
$\langle bs53 \rangle$	$::=$	$\langle binary \rangle$	(394)
		$ \langle binary \rangle \langle bs54 \rangle$	(395)
$\langle bs54 \rangle$	$::=$	$\langle binary \rangle$	(396)
		$ \langle binary \rangle \langle bs55 \rangle$	(397)
$\langle bs55 \rangle$	$::=$	$\langle binary \rangle$	(398)
		$ \langle binary \rangle \langle bs56 \rangle$	(399)
$\langle bs56 \rangle$	$::=$	$\langle binary \rangle$	(400)
		$ \langle binary \rangle \langle bs57 \rangle$	(401)
$\langle bs57 \rangle$	$::=$	$\langle binary \rangle$	(402)
		$ \langle binary \rangle \langle bs58 \rangle$	(403)
$\langle bs58 \rangle$	$::=$	$\langle binary \rangle$	(404)
		$ \langle binary \rangle \langle bs59 \rangle$	(405)
$\langle bs59 \rangle$	$::=$	$\langle binary \rangle$	(406)
		$ \langle binary \rangle \langle bs60 \rangle$	(407)
$\langle bs60 \rangle$	$::=$	$\langle binary \rangle$	(408)
		$ \langle binary \rangle \langle bs61 \rangle$	(409)
$\langle bs61 \rangle$	$::=$	$\langle binary \rangle$	(410)
		$ \langle binary \rangle \langle bs62 \rangle$	(411)
$\langle bs62 \rangle$	$::=$	$\langle binary \rangle$	(412)
		$ \langle binary \rangle \langle bs63 \rangle$	(413)
$\langle bs63 \rangle$	$::=$	$\langle binary \rangle$	(414)
			(415)